

Statistics/Data analysis^(R)

```
User: models1to4
```

Statistics/Data analysis (R) 16.1

Special Edition

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. Maximum number of variables is set to 5,000; see [help set maxvar](#).

Running C:\Program Files\Stata16\sysprofile.do ...

```
1 . import delimited "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\~all_cog_var_pivot_df.csv"
  (20 vars, 154,340 obs)

2 . mixed score c.ageminus50##(1.smoke 1.diabetes 1.blood_pressure 1.psych_problems 1.doesnt_drink 1.underweight) hispani
  > arly [pweight = stdweights] || id: ageminus50
```

Obtaining starting values by EM:

Performing gradient-based optimization:

```
Iteration 0: log pseudolikelihood = -387049.73
Iteration 1: log pseudolikelihood = -387044.52
Iteration 2: log pseudolikelihood = -387044.52
```

Computing standard errors:

```
Mixed-effects regression      Number of obs   =   154,340
Group variable: id           Number of groups =    30,931
```

```
Obs per group:
      min =      1
      avg =     5.0
      max =      9
```

Log pseudolikelihood = -387044.52	Wald chi2(24)	=	16873.10
	Prob > chi2	=	0.0000

(Std. Err. adjusted for 30,931 clusters in id)

score	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
ageminus50	-.0672501	.0029728	-22.62	0.000	-.0730766	-.0614235
1.smoke	-.6293703	.0697481	-9.02	0.000	-.766074	-.4926666
1.diabetes	.0509591	.073836	0.69	0.490	-.0937567	.1956749
1.blood_pressure	.0298941	.0550381	0.54	0.587	-.0779786	.1377669
1.psych_problems	-.1128022	.0761958	-1.48	0.139	-.2621432	.0365389
1.doesnt_drink	.0951043	.0494033	1.93	0.054	-.0017244	.191933
1.underweight	-.0781041	.1977595	-0.39	0.693	-.4657055	.3094974
smoke#c.ageminus50						
1	.0298719	.0046316	6.45	0.000	.0207941	.0389497
diabetes#c.ageminus50						
1	-.014233	.0037628	-3.78	0.000	-.021608	-.006858

blood_pressure#c.ageminus50							
1	-.0232283	.0029341	-7.92	0.000	-.028979	-.0174775	
psych_problems#c.ageminus50							
1	-.0186752	.0039417	-4.74	0.000	-.0264009	-.0109495	
doesnt_drink#c.ageminus50							
1	-.0215688	.0025724	-8.38	0.000	-.0266105	-.0165271	
underweight#c.ageminus50							
1	-.0260699	.0079106	-3.30	0.001	-.0415744	-.0105654	
hispanic	-.8547275	.0608536	-14.05	0.000	-.9739985	-.7354566	
is_depressed	-.389815	.0279784	-13.93	0.000	-.4446516	-.3349784	
education_level							
1	2.907519	.0578852	50.23	0.000	2.794066	3.020972	
2	4.361999	.066514	65.58	0.000	4.231634	4.492364	
3	5.252202	.0789929	66.49	0.000	5.097379	5.407025	
exercise_freq							
1	-.1621771	.030934	-5.24	0.000	-.2228066	-.1015476	
2	-.8034224	.0415203	-19.35	0.000	-.8848007	-.722044	
heavy_drinker	-.0973818	.0562122	-1.73	0.083	-.2075557	.0127921	
marital_end	-.2445092	.0342988	-7.13	0.000	-.3117337	-.1772847	
trouble_falling	-.1077382	.0360629	-2.99	0.003	-.1784201	-.0370563	
wake_up_early	-.0286562	.0343286	-0.83	0.404	-.0959391	.0386267	
_cons	14.41603	.0808286	178.35	0.000	14.25761	14.57445	

Random-effects Parameters	Estimate	Robust Std. Err.	[95% Conf. Interval]	
id: Independent				
var(ageminus50)	.0095152	.0003345	.0088816	.010194
var(_cons)	5.663681	.0910585	5.487992	5.844994
var(Residual)	6.175168	.0495401	6.078831	6.273032

Warning: Sampling weights were specified only at the first level in a multilevel model. If these weights are indicative

```
3 . save "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta"
file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta saved
```

```
4 . estimates store model1
```

```
5 . mixed score ageminus50 smoke diabetes blood_pressure psych_problems doesnt_drink underweight hispanic is_depressed i.
> tdweights] || id:
variable falling not found
r(111);
```

```
6 . mixed score ageminus50 smoke diabetes blood_pressure psych_problems doesnt_drink underweight hispanic is_depressed i
> stdweights] || id:
```

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0: log pseudolikelihood = -388835.34

Iteration 1: log pseudolikelihood = -388835.34

Computing standard errors:

Mixed-effects regression Number of obs = 154,340
Group variable: id Number of groups = 30,931

Obs per group:
 min = 1
 avg = 5.0
 max = 9

Log pseudolikelihood = -388835.34 Wald chi2(18) = 17348.16
 Prob > chi2 = 0.0000

(Std. Err. adjusted for 30,931 clusters in id)

score	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
ageminus50	-.098074	.0018309	-53.57	0.000	-.1016625	-.0944856
smoke	-.2582396	.0454153	-5.69	0.000	-.3472519	-.1692273
diabetes	-.1224524	.0396506	-3.09	0.002	-.2001661	-.0447387
blood_pressure	-.1816628	.0336524	-5.40	0.000	-.2476203	-.1157052
psych_problems	-.3333113	.0484013	-6.89	0.000	-.4281761	-.2384465
doesnt_drink	-.2586301	.0293696	-8.81	0.000	-.3161936	-.2010667
underweight	-.6963966	.1015288	-6.86	0.000	-.8953893	-.4974038
hispanic	-.7241297	.0596791	-12.13	0.000	-.8410987	-.6071608
is_depressed	-.4127532	.0286919	-14.39	0.000	-.4689884	-.356518
education_level						
1	2.983992	.0556102	53.66	0.000	2.874998	3.092986
2	4.46819	.0645689	69.20	0.000	4.341637	4.594743
3	5.280905	.0779998	67.70	0.000	5.128028	5.433782
exercise_freq						
1	-.1891593	.0322551	-5.86	0.000	-.2523781	-.1259405
2	-.9322027	.0433232	-21.52	0.000	-1.017115	-.8472908
heavy_drinker	-.1305415	.0586995	-2.22	0.026	-.2455904	-.0154927
marital_end	-.2785924	.0349642	-7.97	0.000	-.3471209	-.2100638
trouble_falling	-.0777943	.0378762	-2.05	0.040	-.1520302	-.0035584
wake_up_early	-.017649	.0352783	-0.50	0.617	-.0867932	.0514953
_cons	14.67407	.0748824	195.96	0.000	14.5273	14.82083

Random-effects Parameters		Estimate	Robust Std. Err.	[95% Conf. Interval]	
id: Identity					
	var(_cons)	7.322957	.087097	7.154224	7.495669
	var(Residual)	6.668155	.0519929	6.567026	6.770842

Warning: Sampling weights were specified only at the first level in a multilevel model. If these weights are indicative

7 . estimates store model2

8 . estimates stats model1 model2

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
<u>model1</u>	154,340	.	-387044.5	28	774145	774423.6
<u>model2</u>	154,340	.	-388835.3	21	777712.7	777921.6

Note: BIC uses N = number of observations. See [\[R\] BIC note](#).

9 . save "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta", replace
file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta saved

10 . mixed score c.ageminus50##(1.smoke) diabetes blood_pressure psych_problems doesnt_drink underweight hispanic is_depres

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0: log pseudolikelihood = **-388810.86**

Iteration 1: log pseudolikelihood = **-388810.86**

Computing standard errors:

Mixed-effects regression Number of obs = **154,340**
Group variable: id Number of groups = **30,931**

Obs per group:

min = **1**
avg = **5.0**
max = **9**

Log pseudolikelihood = **-388810.86** Wald chi2(17) = **17389.10**
Prob > chi2 = **0.0000**

(Std. Err. adjusted for **30,931** clusters in id)

score	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
ageminus50	-.1006457	.0018909	-53.23	0.000	-.1043517	-.0969396
1.smoke	-.5805837	.0742088	-7.82	0.000	-.7260304	-.4351371
smoke#c.ageminus50						
1	.026956	.0047947	5.62	0.000	.0175586	.0363534
diabetes	-.1241512	.0396599	-3.13	0.002	-.2018832	-.0464191
blood_pressure	-.1856481	.0336823	-5.51	0.000	-.2516641	-.119632
psych_problems	-.343166	.0481604	-7.13	0.000	-.4375586	-.2487734
doesnt_drink	-.2609896	.0293738	-8.89	0.000	-.3185611	-.2034181
underweight	-.7161723	.1012885	-7.07	0.000	-.914694	-.5176505
hispanic	-.7308724	.0597085	-12.24	0.000	-.847899	-.6138458
is_depressed	-.4179097	.0284191	-14.71	0.000	-.4736101	-.3622093
education_level						
1	2.988898	.0555651	53.79	0.000	2.879992	3.097803
2	4.47086	.0644657	69.35	0.000	4.344509	4.59721
3	5.283141	.0779172	67.80	0.000	5.130426	5.435856
exercise_freq						
1	-.189846	.032241	-5.89	0.000	-.2530373	-.1266548
2	-.9332867	.0433088	-21.55	0.000	-1.01817	-.8484031

(Std. Err. adjusted for 30,931 clusters in id)

score	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]	
ageminus50	-.100956	.0018771	-53.78	0.000	-.104635	-.0972769
1.smoke	-.6613908	.0693662	-9.53	0.000	-.7973461	-.5254355
smoke#c.ageminus50						
1	.033264	.0046166	7.21	0.000	.0242157	.0423123
diabetes	-.1757474	.0391059	-4.49	0.000	-.2523935	-.0991012
blood_pressure	-.2847776	.0325546	-8.75	0.000	-.3485834	-.2209717
psych_problems	-.3855823	.0469417	-8.21	0.000	-.4775863	-.2935782
doesnt_drink	-.2184776	.0285428	-7.65	0.000	-.2744205	-.1625348
underweight	-.6532772	.0967727	-6.75	0.000	-.8429482	-.4636062
hispanic	-.8535544	.0607192	-14.06	0.000	-.9725619	-.7345469
is_depressed	-.4007042	.0277988	-14.41	0.000	-.4551888	-.3462196
education_level						
1	2.912944	.0577946	50.40	0.000	2.799669	3.026219
2	4.350526	.0663398	65.58	0.000	4.220502	4.480549
3	5.245115	.0788485	66.52	0.000	5.090575	5.399655
exercise_freq						
1	-.166843	.0309398	-5.39	0.000	-.2274838	-.1062022
2	-.8316632	.0415551	-20.01	0.000	-.9131098	-.7502167
heavy_drinker	-.1320735	.0561827	-2.35	0.019	-.2421896	-.0219574
marital_end	-.2658612	.0342809	-7.76	0.000	-.3330506	-.1986718
_cons	14.86494	.0745845	199.30	0.000	14.71876	15.01112

Random-effects Parameters	Estimate	Robust Std. Err.	[95% Conf. Interval]	
id: Independent				
var(ageminus50)	.0096532	.0003364	.0090158	.0103355
var(_cons)	5.613025	.0898954	5.43957	5.792011
var(Residual)	6.193226	.0497333	6.096513	6.291472

Warning: Sampling weights were specified only at the first level in a multilevel model. If these weights are indicative

14 . estimates store model4

15 . estimates stats model1 model2 model3 model4

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
<u>model1</u>	154,340	.	-387044.5	28	774145	774423.6
<u>model2</u>	154,340	.	-388835.3	21	777712.7	777921.6
<u>model3</u>	154,340	.	-388810.9	20	777661.7	777860.7
<u>model4</u>	154,340	.	-387227	21	774496	774704.9

Note: BIC uses N = number of observations. See [\[R\] BIC note](#).

```

16 . save "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta", replace
    file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta saved

17 . scalar LR_obs = 2*(e(ll_model4)-e(ll_model3))
    unknown function 2()
    r(133);

18 . scalar LR_obs = 2*(e(ll_model4)-e(ll_model3))

19 . predict r, resid

20 . predict yhat, xb

21 . scatter r yhat

22 . save "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta", replace
    file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Mar7th.dta saved

23 . graph export "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\model4residvsfitted.jpg", as(jpg) name("G
    (file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\model4residvsfitted.jpg written in JPEG format)

24 . qnorm r

25 . graph export "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Model4qnorm.jpg", as(jpg) name("Graph") q
    (file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Model4qnorm.jpg written in JPEG format)

26 . predict u, reffects
    too few variables specified
    you must specify 2 new variable(s)
    r(102);

27 . scatter r ageminus50

28 . graph export "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\model4scatterresidtime.jpg", as(jpg) name
    (file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\model4scatterresidtime.jpg written in JPEG format)

29 . predict u0 u1, reffects

30 . qnorm u0

31 . graph export "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\model4qnormrandint.jpg", as(jpg) name("Gra
    (file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\model4qnormrandint.jpg written in JPEG format)

32 . qnorm u1

33 . graph export "C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Model4qnormrandslope.jpg", as(jpg) name("C
    (file C:\Users\fari9860\OneDrive - Wilfrid Laurier University\MA489\Model4qnormrandslope.jpg written in JPEG format)

34 . estimates save models1to4
    file models1to4.ster saved

35 .

```